

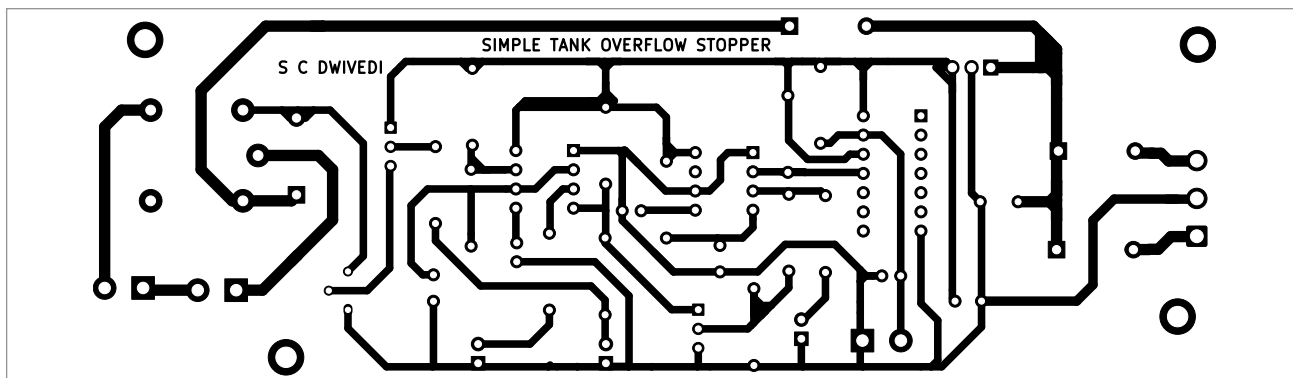


Fig. 6: Actual-size PCB

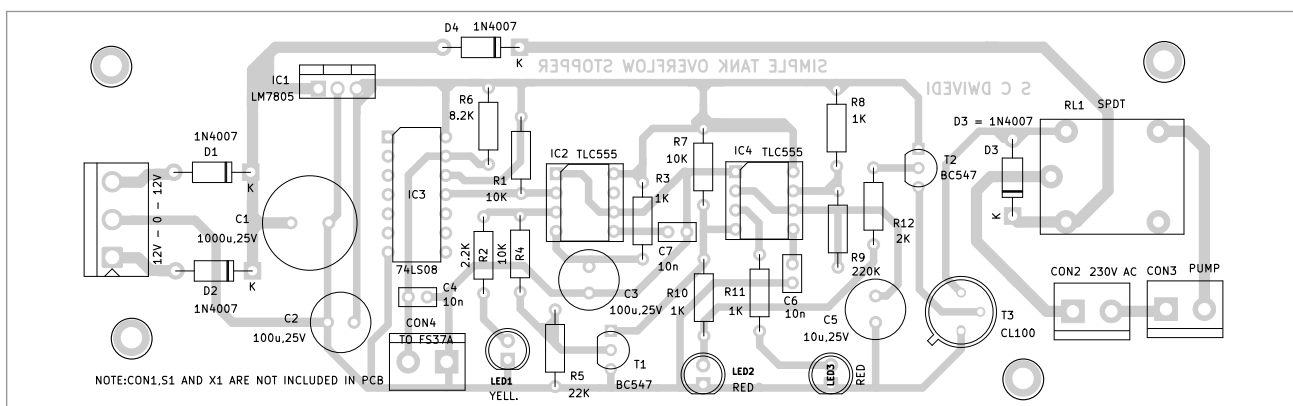


Fig. 7: Component layout

the stagnant water level has fallen sufficiently to bring the float back to its initial resting position on the stopper, restoring the reed switch to its open state.

The overflow stopper system requires two separate power supplies: one for the relay-control electronics and another for the relay coil. The control electronics operate on a regulated +5V supply, while the relay requires a +12V supply to ensure reliable operation.

Construction and testing

An actual-size, single-side PCB layout for the tank overflow stopper system (Fig. 5) is shown in Fig. 6, along with its component layout in Fig. 7. Once the circuit is assembled on the PCB, it should be enclosed in a suitable cabinet. The footprints for CON1, S1, and the transformer are not included on the PCB; therefore, these components must be mounted externally.

PARTS LIST

Semiconductors:

IC1	- LM7805, 5V regulator
IC2	- TLC555 timer
IC3 (A-E)	- 74LS08 quadruple two-input AND gate
D-D4	- 1N4007 rectifier diode
T1, T2	- BC547 NPN transistor
T3	- CL100 NPN transistor
LED1, LED2	- 5mm yellow, red LEDs
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1, R4, R7	- 10k Ω
R2	- 2.2k Ω
R3, R8, R10, R11	- 1k Ω
R5	- 22k Ω
R6	- 8.2k Ω
R9	- 220k Ω
R12	- 2k Ω

Capacitors:

C1	- 1000 μ F, 25V electrolytic
C2, C3	- 100 μ F, 25V electrolytic
C4, C6, C7	- 10nF ceramic disc
C5	- 10 μ F, 25V electrolytic

Miscellaneous:

CON1-CON4	- 2-pin connector
SENSOR	- FS37A float sensor
S1	- On/off switch
RL1	- 12V DPDT relay (PCB mounted)
X1	- 230V AC primary to 12V-0-12V AC, 500mA secondary transformer

EFY note: Only IC3D and IC3E are used here.

Alternatively, the system can be assembled and tested on three separate Veroboard: one for the power supply unit, one for the relay-driving electronics, and the third for the relay itself. Mounting the relay separately enhances safety because the 230V AC mains enters the PCB through the relay contacts. All three boards, along with the transformer, can be housed in a metal enclosure. Wires from the sensor assembly may be connected to the circuit board through a 2-pin electrical connector.

Standard PVC electrical wires can be used for interconnections and to connect the sensor to the electronics board. Push-type 6mm thimbles are helpful to connect the relay contacts to the pump motor. **EFY**



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